

CSE400 – Fundamentals of Probability in Computing

Lecture 3: Introduction to Probability Theory

Lecture Scribe for Exam Revision

1 Opening Motivation (Page 2)

The lecture begins with the quotation:

“Good is the enemy of great. That’s why so few things become great.” –
James C. Collins

Step-by-Step Meaning

- “Good” means doing only basic or minimum work.
- “Great” means deep understanding and continuous improvement.
- If students stop at “good enough,” they will not master concepts.
- In CSE400, students are expected to aim for strong conceptual clarity.
- This quotation prepares students mentally for serious learning.

2 Learning Attitude: Growth and Fixed Mindset (Pages 3–4)

2.1 Growth Mindset

The lecture presents the following ideas:

- “Failure is an opportunity to grow”
- “I like to try new things”
- “I can learn to do anything I want”

- “Challenges help me grow”
- “My effort and attitude determine my abilities”
- “Feedback is constructive”
- “I am inspired by the success of others”

Step-by-Step Understanding

- Failure is treated as a learning step.
- Trying new methods is encouraged.
- Learning is possible through effort.
- Difficult problems help improvement.
- Ability is linked to practice.
- Feedback helps correction.
- Others’ success motivates learning.

So, students are expected to:

- Practice regularly.
- Accept mistakes.
- Improve continuously.

2.2 Fixed Mindset

The lecture contrasts this with:

- “Failure is the limit of my abilities”
- “My abilities are unchanging”
- “I don’t like to be challenged”
- “My potential is predetermined”
- “When I’m frustrated, I give up”
- “I stick to what I know”

Step-by-Step Understanding

- Failure is seen as permanent.
- Learning is considered limited.
- Difficult topics are avoided.
- Effort is reduced.

2.3 Conclusion from Comparison

- Probability needs practice and patience.
- A growth mindset helps understanding.
- A fixed mindset blocks learning.
- Therefore, students should follow a growth mindset in CSE400.

3 Why Should We Learn CSE400? (Pages 6–8)

The outline introduces:

- “Why should we learn CSE400?”
- Example: Daily life conversations
- Engineering Applications

3.1 Daily Life Conversations

The lecture connects probability with daily thinking.

Classroom-Based Understanding

In daily life, people say:

- “Maybe it will rain.”
- “There is a chance.”
- “Probably it will happen.”

Step-by-Step Explanation

- Many situations are uncertain.
- People estimate chances informally.
- They use words like “maybe” and “probably”.
- These are informal probabilities.
- CSE400 teaches how to represent these ideas mathematically.

So:

Informal thinking $\rightarrow$ Formal probability models.

3.2 Engineering Applications

The lecture lists three main areas.

3.2.1 (a) Speech Recognition

- Speech signals contain noise.
- Words may not be clear.
- System receives uncertain input.
- Multiple interpretations are possible.
- Probability helps choose the most likely word.

3.2.2 (b) System Radar System

- Radar sends signals.
- Signals return with noise.
- Objects may or may not be present.
- Detection is uncertain.
- Probability decides whether a target exists.

3.2.3 (c) Communication Network

- Data is transmitted in packets.
- Errors and delays occur.
- Transmission is uncertain.
- Probability analyzes performance.
- Systems are optimized using probabilistic models.

3.3 Conclusion

- Engineering systems face randomness.
- Exact prediction is not possible.
- Probability provides tools to handle uncertainty.
- Therefore, CSE400 is essential.

4 Project Component in CSE400 (Pages 24–26)

4.1 Project Weightage

The lecture states:

CSE400 – Project Kickoff (30%)

So:

- Project contributes 30% to evaluation.

4.2 Team Formation (Page 25)

Deadline: 17th January 2026, Saturday – EOD

Meaning:

- Teams must be formed before deadline.
- Submission is time-bound.
- Planning must start early.

4.3 Project Execution Guidelines (Page 26)

Main Components:

- Concept Evolution Maps
- Scribe: Process & Decision-Making
- Multimodal Artifacts
- Question-Driven Artifacts
- Collaboration & Team Dynamics
- Deliverables
- Project Viva

Step-by-Step Meaning

- Work is divided into stages.
- Each stage has specific outputs.
- Students must document thinking.
- Evaluation is continuous.
- Final understanding is tested in viva.

5 Project Execution: Milestones (Page 27)

5.1 M1: Kickstart

Includes:

- Team Formation
- Area Identification
- Background
- Motivation
- Problem Formulation

5.2 M2: Mathematical Modeling

Includes:

- Random Variables (RV)
- PMF / PDF
- CDF
- Multivariate RV
- Joint PMF / PDF / CDF

5.3 M3: Coding

Includes:

- Simulation
- Computation

5.4 M4: Inference

Includes:

- Choose randomized algorithm
- Understand it
- Code it

5.5 M5: Randomized Algorithms

Includes:

- Apply algorithm
- Compare with deterministic
- Analyze results

5.6 M6: Derivation and Analysis

Includes:

- Derive bounds
- Analysis
- Final submission

6 Submission #1: Concept Evolution Maps (Pages 28–29)

Tools:

- miro.com
- diagrams.net

Purpose:

- Show concept connections.
- Organize ideas visually.
- Explain flow of learning.

7 Submission #2: Scribe – Learning Reflection Logs (Page 30)

Types:

- Lecture Scribes
- Project Scribes

Must include:

- Decision logs
- Constraints
- Alternatives
- Evidence
- Trade-offs

8 Submission #3: Multimodal Artifacts (Page 31)

Includes:

- Video / Audio / Visual
- Duration: 10–15 minutes

- Use PPT/Slides
- Think-Aloud Videos
- One-Minute Insights
- Project Demo

9 Classroom Interaction and Dialogue

9.1 Active Learning (Pages 13–15)

- Active learning
- Class discussion
- Campuswire

9.2 Q&A Session (Page 87)

“Q & A Session: Open Discussions”

10 Integrated Understanding of CSE400

10.1 Learning Philosophy

- Growth mindset
- Reflection
- Discussion
- Practice

10.2 Technical Focus

- Random variables
- Distributions
- Simulation
- Randomized algorithms
- Analysis

10.3 Practical Focus

- Projects
- Documentation
- Videos
- Presentations

10.4 Dialogue-Based Learning

Learning happens through:

- Daily examples
- Classroom discussion
- Q&A sessions
- Reflection logs

Final Summary for Exam Revision

- CSE400 promotes a growth mindset.
- Probability formalizes uncertainty.
- Daily life uses informal probability.
- Engineering systems depend on probabilistic models.
- Project weightage is 30%.
- Work is divided into M1–M6.
- Mathematical modeling is central.
- Scribes explain reasoning.
- Videos show thinking.
- Open discussions support learning.